# Kevin Nam (남기빈)

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## Contents

## About Me

I am a Ph.D. candidate in the Security Optimization Research (SOR) Lab at Seoul National University. My research trajectory began with hardware security topics, including monitoring mechanisms and Trusted Execution Environments (TEE), and has since expanded to privacy-enhancing technologies (PETs). In particular, my most recent works focused on on enabling efficient and accurate Machine Learning as a Service over Fully Homomorphic Encryption (FHE).

My research vision is to ***design efficient Security and Privacy Enhancing Computing Systems***. While PETs have achieved significant theoretical and mathematical progress, they are still far from widespread deployment due to their computational complexity. I draw inspiration from the history of Artificial Intelligence: envisioned since the 1970s, AI only became practically deployable with 21st-century hardware and software innovations such as GPUs, distributed systems, and large-scale frameworks. I believe PETs can follow a similar path, becoming practical through retrofitted HW/SW innovations, which is the central goal of my research.

To this end, I pursue what I call non-mathematical approaches-—hardware acceleration, compiler optimization, and algorithm–system co-design–to mitigate the performance bottlenecks of PETs, including Homomorphic Encryption, Secure Multi-Party Computation, and TEEs. Ultimately, my goal is to make privacy-preserving computation as conventional and accessible as everyday computing.

## Research Interests

- Applied Cryptography
- Systems (HW/SW) for PETs & Cryptography
- HW/SW Security (including confidential computing)
- Security/Privacy for AI Services
- Compiler
- Computer architecture

## Education

**Seoul National University**, MS/Ph.D in Electrical and Computer Engineering — Mar 2020 – Feb 2026 (expected)

Advisor : Prof. Yunheung Paek
Dissertation Title (tentative) : Non-Mathematical Optimizations for Fast and Precise Privacy-Preserving Machine Learning System over Encrypted Data

**Seoul National University**, BS in Electrical and Computer Engineering — Mar 2014 – Feb 2020

Left for military service (ROKAF) during 2017-2019

## Research Experience

**Seoul National University.** Seoul, Korea — Mar 2020 – Feb 2026*

Research Assistant
Advisor: Yunheung Paek

**Yale University.** New Haven, CT, USA — Jan 2023 – Feb 2023

Visiting Student
Advisor: Jakub Szefer
**Post-Quantum-Cryptography Accelerator Design, Side-Channel Attacks on FPGA**

**Seoul National University.** Seoul, Korea — Jun 2018 – Dec 2018

Intern
Advisor: Yunheung Paek
**HW Implementation of LSTM based Anomalous Branch Behavior Monitoring**

## Publications

### Refereed Conference Publications

[1] [HiPC‘25] " An Accelerator for Low-computational Overhead Privacy-Preserving GNN Inference". *Heonhui Jung*, Whoiree Ha*, Kevin Nam, Youyeon Joo, Lucas Oros, and Yunheung Paek. (* co-first authors) In the 32nd IEEE International Conference on High Performance Computing, Data, and Analytics (HiPC‘25), December 2025.*
(BK21 IF: 1, CSRankings).

[2] **[Security‘25]** " SLOTHE: Lazy Approximation of Non-Arithmetic Neural Network Functions over Encrypted Data". *Kevin Nam*, Youyeon Joo*, Seungjin Ha, and Yunheung Paek. (* co-first authors) In USENIX Security Symposium (Security‘25), August 2025.*
**(KIISE S, BK21 IF: 3, CSRankings).**

[3] **[ASPLOS‘25]** "Affinity-based Optimizations for TFHE on Processing-in-DRAM". *Kevin Nam, Heonhui Jung, Hyunyoung Oh, and Yunheung Paek. In the 30th International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS‘25), April 2025.*
**(KIISE S, BK21 IF: 4, CSRankings).**

[4] **[Security‘25]** "LOHEN: Layer-wise Optimizations for Neural Network Inferences over Encrypted Data with High Performance or Accuracy". *Kevin Nam, Youyeon Joo, Dongju Lee, Seungjin Ha, Hyunyoung Oh, Hyungon Moon, and Yunheung Paek. In USENIX Security Symposium (Security‘25), August 2025.*
**(KIISE S, BK21 IF: 3, CSRankings).**

[5] **[ICCAD‘22]** "Accelerating N-bit Operations over TFHE on Commodity CPU-FPGA". *Kevin Nam, Hyunyoung Oh, Hyungon Moon, and Yunheung Paek. In IEEE/ACM International Conference on Computer-Aided Design (ICCAD‘22), November 2022.*
**(KIISE A, BK21 IF: 3, CSRankings).**

[6] "Area-Efficient Accelerator for the Full NTRU-KEM Algorithm". *Yongseok Lee, Kevin Nam, Youyeon Joo, Jeehwan Kim, Hyunyoung Oh, Yunheung Paek. In International Conference on Computational Science and Its Applications (ICCSA‘23). Cham: Springer Nature Switzerland, 2023.*

### Refereed Journal Publications

[7] "An Efficient Hardware/Software Co-design for FALCON on Low-End Embedded Systems". *Yongseok Lee, Jonghee Youn, Kevin Nam, Heon Hui Jung, Myunghyun Cho, Jimyung Na, Jong-Yeon Park, Seungsu Jeon, Bo Gyeong Kang, Hyunyoung Oh, Yunheung Paek. In IEEE Access, 2024.*

[8] "MeetGo: A trusted execution environment for remote applications on FPGA". *Hyunyoung Oh, Kevin Nam, Seongil Jeon, and Yeongpil Cho, Yunheung Paek. In IEEE Access, 2021.*

### Refereed Poster Publications

[9] "Affinity-based Optimizations of Homomorphic Encryption Operations on Processing-in-DRAM". *Kevin Nam, Heonhui Jung, Hyunyoung Oh, and Yunheung Paek. In the 61th ACM/IEEE Design Automation Conference Work-in-Progress (DAC‘24 WiP), 2024.*

[10] "Implementing Efficient, Precise N-bit Operations of TFHE on Commodity CPU-FPGA". *Kevin Nam, Youyeon Joo, Dongju Lee, Seungjin Ha, Hyunyoung Oh, Hyungon Moon, and Yunheung Paek. In the 59th ACM/IEEE Design Automation Conference Work-in-Progress (DAC‘22 WiP), 2022.*

### Domestic Papers *(first author ones only)*

[11] "Quantization of Non-Arithmetics to Optimize CNNs over TFHE". *Kevin Nam, Heonhui Jung, Dongju Lee, Yunheung Paek. Annual Symposium of KIPS, 2024*

[12] "Intermediate Data Guided Approximation for Fast and Accurate Encrypted Neural Networks". *Kevin Nam, Youyeon Joo, Seungjin Ha, Yunheung Paek. Annual Conference of KIPS, 2024.*

[13] "Side-Channel Attacks on Homomorphic Encryption and Their Mitigation Methods". *Kevin Nam, Youyeon Joo, Yunheung Paek. Annual Symposium of KIPS, 2023* **(NIPA Director Award)**

[14] "Realization of Homomorphic Encrypted Deep Learning Models". *Kevin Nam, Myunghyun Cho, Hyunjun Kim, Yunheung Paek. Annual Conference of KIPS, 2021,* **(Undang Academic Award)**

[15] "Partially Homomorphic Encryption HW accelerator". *Kevin Nam, Jiwon Chang, Myunghyun Cho, Inyoung Bang, Yunheung Paek. Annual Conference of KIPS, 2020*

#### Patents

[16] "METHOD FOR PROCESSING HOMOMORPHIC CIPHIERTEXT AND ELECTRONIC APPARATUS". *Yunheung Paek, Heonhui Jung, Kevin Nam, Seungjin Ha, Junbum Shin, Inkwan Yu, Sunchul Jung, Jungjoo Seo. Feb 2025. US Patent No. 19/052442, Applied*

[17] "METHOD FOR PROCESSING HOMOMORPHIC CIPHIERTEXT AND ELECTRONIC APPARATUS". *Yunheung Paek, Heonhui Jung, Kevin Nam, Seungjin Ha, Junbum Shin, Inkwan Yu, Sunchul Jung, Jungjoo Seo. Jan 2025. KR Patent No. 10-2025-0008424, Applied*

[18] "METHOD AND APPRATUS FOR COMPUTING ENCRYPTED DATA USING MULTI-HOMOMORPHIC ENCRYPTION SYSTEM". *Yunheung Paek, Kevin Nam, Youyeon Joo. Nov 2024. KR Patent No. 10-2024-0153707, Applied*

[19] "TEE ENVIRONMENT PROVIDING APPARATUS AND METHOD USING FPGA". *Yunheung Paek, Hyunyoung Oh, Kevin Nam, Yeongpil Cho. Jan 2021. KR Patent No. 10-2021-0006281, Applied*

## Research Projects

*w. fundings, Principal Investigator : Yunheung Paek.*

**Privacy-Preserving Neural Network Inference System** *funded by NRF, South Korea* Oct 2023 – current
- Integrate non-mathematical methods (e.g., compiler) to enhance the performance of cryptographic backends

**Secure FHE Key Management System** *funded by CryptoLab, South Korea* Mar 2023 – April 2024
- Designed efficient TEE-based FHE key management system within SGX restrictions

**Neural Network Program to FHE Transpiler** *funded by ETRI, South Korea* Oct 2023 – Nov 2023
- Designed a transpiler that translates tensorflow/Pytorch programs into FHE programs

**PEC Platform** *funded by the National Intelligence Service, South Korea* Mar 2022 – Nov 2023
- Developed a framework that analyzes normal codes and suggest appropriate PETs and generates a secure code

**FPGA Accelerator for CKKS** *funded by CryptoLab, South Korea* Mar 2022 – Nov 2022
- Participated as the algorithm analyzer - designed the overall operation mapping and pipeline for CKKS on FPGA

**FPGA Accelerator for TFHE** *funded by the National Intelligence Service* Mar 2021 – Nov 2021
- Designed an FPGA-based TFHE accelerator exploiting the double parallelism of TFHE

**Designing HW/SW-based Computing Base for Secure Boxes** *funded by IITP, South Korea* Sep 2021 – current
- Implemented HW/SW-based isolated VMs with TEEs and HE

**Designing PQC Accelerators** *funded by Samsung Electronics, South Korea* Sep 2020 – Sep 2024
- Integrated and optimized the HW modules

**FPGA-based Trusted Execution Environments** *funded by NRF, South Korea* Mar 2020 – July 2022
- Implemented TEE isolation systems on FPGAs

## Teaching

**Seoul National University**
- (Teaching Assistant) Logic Circuit Design Course, 430.201A
  Mar 2020 - Aug 2020, Sep 2022 - Dec 2022, Sep 2023 - Dec 2023, Sep 2024 - Dec 2024
- (Teaching Assistant) Data Security & Privacy, 430.658
  Sep 2024 - Dec 2024, Sep 2025 - Dec 2025
- (Teaching Assistant) Creative Design Project, 430.405
  Mar 2022 - current

## Honors & Awards

**Excellence Paper Award**, *from ACK 2024* May 2024

**Award for Excellence in Teaching Assistant**, *from Seoul National University* Jan 2024
*Logic circuit Design Course*

**NIPA Director Award** *from ASK 2023* Jun 2023

**Undang Academic Award** *from KIPS, Best Graduate Student Paper Award* Dec 2021

**Best Paper Award** *from ASK 2021* May 2021

**Scholarship**
BK 21+ Scholarship *by the Ministry of Education of Korea* Sep 2020 - Feb 2021
Mar 2023 - present

## Services & Activities

**IEEE Symposium on Security and Privacy 2026** *Artifact Evaluation Committee*

**EuroSys 2025** *Shadow Program Committee*

**CES 2017** *Exihibitor (BOM Team)*

## Invited Talks

[1] "*Zero-Error Privacy-Preserving Machine Learning as a Service*" *presented at Seoul National University AI Health & Care Center Symposium, December 2023*

[2] "*Fully Homomorphic Encryption based Neural Network Inference*" *presented at Ewha Women University, September 2022*